=== UNIT 3: TENSES, ASPECT & CONDITIONAL CLAUSE SYSTEMS ===

>> 3.1 The Twelve Tense-Aspect System in Technical Communication

Scientific writing demands precise tense-aspect selection to reflect the chronological status and empirical validity of findings.

* Present Tense System:
- Simple Present: Used for universal scientific truths, established natural laws, and general axioms: "Water boils at 100 deg C under atmospheric pressure." Stative verbs (know, believe, consist, contain) cannot be used in continuous aspects (*is containing is ungrammatical).
- Present Continuous: Ongoing dynamic events at the time of speaking: "The temperature is rising steadily."
- Present Perfect: Action initiated in the past with direct relevance or continuation into the present: "Researchers have discovered a novel thermostable Taq polymerase." Uses ’Since’ (specific point in time: since 2018) vs. ’For’ (duration: for five years).
- Present Perfect Continuous: Continuous action extending from past to present: "The culture has been fermenting for 72 hours."

* Past Tense System:
- Simple Past: Completed actions at a specific, stated past time; standard tense for Materials and Methods and Experimental Results sections: "The solution was centrifuged at 10,000 rpm for 15 minutes."
- Past Continuous: Ongoing background action interrupted by another past event: "The bioreactor was operating normally when the power failed."
- Past Perfect: An action completed prior to another past action (anteriority): "The cells had entered apoptosis before the inhibitor was administered."

* Future Tense System: Expressed via modal auxiliaries (will, shall) or periphrastic forms (is going to).

>> 3.2 The Four Conditional Clause Architectures

Conditionals consist of a dependent conditional clause (if-clause / protasis) and an independent consequence clause (main clause / apodosis).

* Zero Conditional (Universal Scientific Truths & Invariant Causality):
- Structure: If + Simple Present, ... Simple Present.
- Example: "If water reaches 100 deg C at 1 atm, it boils." (Invariable thermodynamic law).

* First Conditional (Real / Probable Future Event):
- Structure: If + Simple Present, ... will / can / may + Base Verb (V1).
- Example: "If the fermenter temperature exceeds 40 deg C, the microbial enzymes will denature."

* Second Conditional (Hypothetical / Unreal Present or Future Event):

- Structure: If + Simple Past (Subjunctive 'were' for all persons), ... would / could / might + Base Verb.
- Example: "If the laboratory were equipped with cryogenic electron microscopy, we could resolve the macromolecular structure."

* Third Conditional (Counterfactual Past Regret / Impossible Condition):
- Structure: If + Past Perfect (had + V3), ... would have / could have + Past Participle (V3).
- Example: "If the technician had calibrated the pH electrode yesterday, the entire batch would not have spoiled."

[EXERCISE 1: Tense Consistency in Scientific Research Manuscripts]
Problem: Re-write the following experimental report excerpt, correcting inappropriate shifts between past, present, and perfect tenses:
Inconsistent Draft:
"The researchers synthesize [1] a series of novel bio-polymers. In the current paper, we tested [2] their tensile strength. The results show [3] that polymer A had [4] an ultimate tensile strength of 55 MPa. We conclude [5] that bio-polymers will be [6] viable replacements for petrochemical plastics."
Corrected Academic Version:
"The researchers synthesized [1, Simple Past for completed past synthesis] a series of novel bio-polymers. In the current paper, we report [2, Simple Present for current manuscript contents] their tensile strength. The results show [3, Simple Present for timeless data presentation] that polymer A has [4, Simple Present for established material property] an ultimate tensile strength of 55 MPa. We conclude [5, Simple Present for author deduction] that these bio-polymers are [6, Simple Present for generalized scientific conclusion] viable replacements for petrochemical plastics."

[EXERCISE 2: Conditional Clause Transformation Drill]
Problem: Transform the provided prompt across all four conditional forms, maintaining semantic coherence:
Base Context: Autoclaving at 121 deg C destroys bacterial endospores.
- Zero Conditional: "If an autoclave maintains 121 deg C for 20 minutes, it destroys all bacterial endospores."
- First Conditional: "If the technician runs the autoclave at 121 deg C for 20 minutes, the steam will destroy all bacterial endospores in the media."
- Second Conditional: "If this older autoclave maintained a steady 121 deg C, it would destroy the persistent bacterial endospores."
- Third Conditional: "If the autoclave had maintained 121 deg C for the full 20 minutes, the steam would have destroyed the contaminating endospores."

[EXERCISE 3: Disambiguating 'Since' vs. 'For' in Longitudinal Studies]
Problem: Insert 'since' or 'for' accurately into the following clinical research sentences:
Sentence 1: "The stem cell line has been continuously subcultured in our laboratory [since / for] over six years."
- Solution: FOR. ('over six years' denotes an accumulated duration of time).

Sentence 2: "The biotechnology facility has maintained ISO-9001 compliance [since / for] its inception in March 2015."
- Solution: SINCE. ('March 2015' marks a specific starting point in time).
Sentence 3: "The fermentation broth has been incubating [since / for] 08:00 AM this morning."
- Solution: SINCE. ('08:00 AM' is a defined calendar point of origin).
Sentence 4: "The patient has experienced therapeutic remission [since / for] twelve consecutive months."
- Solution: FOR. ('twelve consecutive months' is a measured temporal span).

[EXERCISE 4: Subjunctive Mood in Hypothetical Conditionals]
Problem: Correct the grammatical error in the following conditional statement, explaining the rule of the Subjunctive Mood:
Faulty Sentence: "If the enzyme concentration was doubled, the initial velocity would increase proportionally."
- Correction: "If the enzyme concentration WERE doubled, the initial velocity would increase proportionally."
- Linguistic Rule: In hypothetical (Second Conditional) constructions expressing unreal or contrary-to-fact scenarios, the past subjunctive form 'were' is mandatory for all grammatical persons (singular and plural: I were, he were, it were). The indicative 'was' is considered informal and ungrammatical in formal scientific prose.

>> 3.3 University Review & Examination Questions

* Q1: Tabulate the four types of conditional clauses in English, showing their if-clause structures, main clause structures, and practical functions with scientific examples.
Answer: (1) Zero Conditional: If-Clause: If + Simple Present; Main Clause: Simple Present. Function: Universal scientific laws and invariant physical facts. Example: 'If heat is applied to ice, it melts.' (2) First Conditional: If-Clause: If + Simple Present; Main Clause: will / can + Base Verb (V1). Function: Real or highly probable future events based on present conditions. Example: 'If the primer concentration is too low, the PCR reaction will fail.' (3) Second Conditional: If-Clause: If + Simple Past (Subjunctive were); Main Clause: would / could + Base Verb (V1). Function: Hypothetical, unlikely, or contrary-to-fact present/future scenarios. Example: 'If the enzyme were thermostable at 95 deg C, it would survive repeated denaturation cycles.' (4) Third Conditional: If-Clause: If + Past Perfect (had + V3); Main Clause: would have + Past Participle (V3). Function: Counterfactual past events that did not happen and cannot be changed. Example: 'If the sample had been stored at -80 deg C, the RNA would not have degraded.'
* Q2: Explain the conventional distribution of tenses across different sections of a standard IMRaD scientific research paper.
Answer: Standard scientific publications adhere to the IMRaD (Introduction, Methods, Results, and Discussion) structural convention: (1) Introduction: Simple Present for

established scientific knowledge and widely accepted theories ('Enzymes catalyze biological reactions'); Present Perfect for surveying past research leading up to current gaps ('Several groups have investigated this gene'). (2) Materials and Methods: Simple Past in passive voice for describing completed laboratory procedures ('The samples were collected, centrifuged, and stored at -20 deg C'). (3) Results: Simple Past for reporting specific empirical findings ('The absorbance increased by 30%'); Simple Present when referring directly to figures and tables ('Figure 2 illustrates the reaction kinetics'). (4) Discussion: Simple Present for interpreting data, stating implications, and formulating conclusions ('These findings demonstrate that the pathway is essential for survival').

>> 3.4 Key Takeaways and Revision Points

* Simple present expresses universal scientific facts; simple past documents completed laboratory procedures.
* Present perfect connects past actions to present outcomes (uses 'since' for point in time, 'for' for duration).
* Zero conditional uses present-present; First conditional uses present-future (will).
* Second conditional uses past subjunctive (were)-would; Third conditional uses past perfect (had)-would have.